

TEJASWI ERATTU TAJ

Seattle, WA | tjerattu@gmail.com | [Portfolio](#) | [LinkedIn](#) | [GitHub](#)

EDUCATION

University of Washington: BS in Informatics
Focus area: Software Engineering + Cyber Security

Expected Graduation: 2028
GPA: 3.8/4.0

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript, HTML/CSS, SQL, PHP, Swift

Frameworks & Tools: React, Node.js, Git, WordPress, MySQL, VS Code Extension API, ARKit, Power BI

Security: Threat Modeling (STRIDE), Vulnerability Assessment, Zero Trust Architecture, IAM/Access Control, Cryptography Fundamentals, Incident Response, Burp Suite, Semgrep

ML/AI: Classification, Regression, Hyperparameter Tuning (scikit-learn, Pandas, NumPy), Computer Vision (YOLOv8, CoreML, Vision), LLM API Integration (OpenAI API)

Cloud/Certs: AWS, Azure, GCP, Docker, CI/CD (GitHub Actions), AWS AI Practitioner, Security+, AWS Cloud Practitioner

PROFESSIONAL EXPERIENCE

Security Engineer, Palana

June 2026 – Present

- Conducted threat modeling across 10+ product surfaces using STRIDE methodology, mapping trust boundaries across the mobile, backend, and Firebase architecture
- Identified a confirmed privilege-escalation vulnerability in session/identity handling and drove remediation with the engineering team prior to deployment
- Evaluated and piloted static analysis tools (Semgrep, npm audit) and ran manual penetration testing with Burp Suite against API/WebSocket endpoints to assess authorization and data-exposure risk

Software Engineering Lead, Accountability & Hopeful Fridays

May 2026 – Present

- Led technical infrastructure planning across 6 systems (GitHub, SharePoint, Microsoft Planner, Trello, WordPress), designing a 6-phase roadmap covering CI/CD, website migration, and member portal
- Designed CI/CD pipeline architecture (feature → development → staging → production) with branch protection and required reviewers for a team of 5+ developers
- Consolidated 40+ cross-team tasks into a centralized tracking system, reducing coordination overhead for a 10+ person organization

Developer Intern, Kerala Association of Washington

May 2025 – Dec 2025

- Migrated 2,800+ member records to WordPress via a custom PHP/MySQL ETL pipeline, cutting manual data entry time by 75%
- Built custom WordPress plugins (PHP, MySQL, JavaScript/AJAX) for real-time member management and automated daily backups via WP-Cron

AI Trainer and Tester, Apollo AI

May 2024 – May 2025

- Trained and evaluated 5 classification/regression models (TensorFlow, scikit-learn) for Saturn AI's grading engine, improving accuracy by 40% for 50+ K-12 students
- Optimized model performance through hyperparameter tuning, cutting average response time from 8s to 3s

Machine Learning Manager, Cyber Minds Non-Profit

Jan 2024 – June 2024

- Built and deployed a Python (scikit-learn) chatbot integrated via Firebase, handling 100+ monthly interactions and cutting response time by 60%
- Led a 5-person team using Agile sprints to integrate a TensorFlow-based ML model into production 2 weeks ahead of schedule

PROJECTS

Findar — AR Object Finder (iOS)

Feb 2026

- Built an AR iOS app (Swift, ARKit, CoreML, YOLOv8n) that locates lost household objects via voice command, detecting 80+ object categories in real time
- Engineered a concurrent on-device detection pipeline (Vision, background queues, frame-skipping) sustaining 60fps while running YOLOv8 inference every 8th frame
- Processed raw LiDAR depth buffers to guide users to lost items via 5-zone directional voice navigation and natural-language command parsing (SFSpeechRecognizer)

GitHub Extension (VS Code)

Jan 2026 – Jun 2026

- Built a TypeScript VS Code extension with a pre-commit secret detection pipeline (regex) scanning staged diffs across 6 credential pattern types
- Designed a real-time Git risk classification engine that scored every action low/medium/high risk, gating destructive operations behind a preview/confirm flow across 6 core workflows

LEADERSHIP AND CREDENTIALS

Languages: English (Fluent), Malayalam (Native), Spanish (Elementary)

Certifications: AWS AI Practitioner, Security+, AWS Cloud Practitioner

Other Leadership: Women in Informatics Finance Director (6 months – present), Palana UW Night Ride Application Security Engineer (3 months – present), Martial Arts Head Instructor (3 years – 2024), Coding Instructor (3 years – 2025)